

Second Entry to the Generalized t -SNE Problem

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1 Current State of the Problem

For the big picture, we aim to establish the convergence theorems for the t -SNE algorithm with generalized input and output kernels. For convenience, if not specified or otherwise explicitly defined, all notations are the same as in [1]. At the currently state, we have shown that the relative entropy t -SNE with generalized kernel g converges and can be computed if g satisfies the following:

$g(y, y') \equiv k(\|y - y'\|)$ such that:

1. $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is decreasing and bounded.
2. g satisfies

$$\iint_{(0, \infty)^2} g(x, x') dx dx' < \infty. \quad (1.1)$$

3. k has bounded derivative.
4. $k'(0) = 0$.

More importantly, the good news is that the proofs concerned with generalized output kernel does not interfere with the input kernel. This significantly reduces the complexity and potential roadblocks for us to continue working on the generalized input kernel.

There are two main steps before we can complete the project. On one hand, we need to give a specific form to the generalized input kernel, then prove the existence and uniqueness of σ^* , because without the constraint on perplexity, t -SNE will fail. On the other hand, we need to prove the bounds for approximation and produce similar results like those in [1]. This document is dedicated to the former, and serves as the second main step in this project, hence the name.

2 The Generalized Input Kernel

The most important question that sets the foundation of the problem is, what does the output kernel look like, given two data points x, x' and parameter σ . We can start by assuming that the input kernel only depends on $\|x - x'\|$ because t -SNE is about the relationship of data points to each other. Then, we may further assume that the input kernel is the function of $\sigma \|x - x'\|$, as σ is used to control the scaling of distance between two data points in the original version. This way we can control the entropy of the induced probability distributions, and it is also the very reason we introduce the concept of perplexity. Finally, because the input kernel is about the probability mass and should always be positive, we can write

it as $\exp(\cdot)$ because the exponential function is a bijection. In particular, we write the input kernel as

$$f_\sigma(x, x') = \exp(-w(\sigma^2 \|x - x'\|^2)), \quad (2.1)$$

where $w : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^+$ is a monotone increasing function. Such expression has several advantages:

1. Since the exponential function is a bijection, we do not lose much generality at all except for how σ interact with the output.
2. The form is in direct comparison with the original kernel. As we dive deeper into the computation we can have clear insights about the constraints we need for the input kernel.
3. The exponential function will cancel out the logarithm in the formula of entropy, greatly simplifying the computation and giving us convenience.
4. The form looks natural and aligns with the original concept and motivation of entropy and perplexity.

With the above being said, we can start adding assumptions to the input kernel. The most immediate and obvious requirements we can impose will be

$$\lim_{t \rightarrow \infty} w(t) = \infty, \quad \int_0^\infty \exp(-w(t)) dt =: Z < \infty. \quad (2.2)$$

These two are the bare minimum for this problem because apparently we want the normalizing quantity to be finite and w must increase to infinity for a decaying tail. Taking $w(t) = t/2$ and we have the original kernel back. We will start with these basic settings and gradually add more requirements as we proceed through the problem. Also, since the input kernel serves as the weight/probability mass, it is always normalized and we can assume $w(0) = 0$ without loss of generality.

3 Existence and Uniqueness of σ^*

We will try to replicate the results in section 2 of [1]. Recall that

$$\begin{aligned} F_{\rho, \mu}(x, \sigma) &= - \int \frac{\exp(-w(\sigma^2 \|x - x'\|^2))}{\int \exp(-w(\sigma^2 \|x - x'\|^2)) d\mu(x')} \log \left(\frac{\exp(-w(\sigma^2 \|x - x'\|^2))}{\int \exp(-w(\sigma^2 \|x - x'\|^2)) d\mu(x')} \right) d\mu(x') + \log \rho \\ &= - \int \frac{w(\sigma^2 \|x - x'\|^2) \exp(-w(\sigma^2 \|x - x'\|^2))}{\int \exp(-w(\sigma^2 \|x - x'\|^2)) d\mu(x')} d\mu(x') \\ &\quad - \log \left(\int \exp(-w(\sigma^2 \|x - x'\|^2)) d\mu(x') \right) + \log \rho. \end{aligned} \quad (3.1)$$

We want to show that $F_{\rho, \mu}(x, \sigma) = 0$ has a unique solution $\sigma^* = \sigma_{\rho, \mu}^*(x)$ for all $\rho \in (0, 1)$ and $x \in \mathbb{R}^d$. First off, we have $\lim_{\sigma \rightarrow 0} F_{\rho, \mu}(x, \sigma) = \log \rho < 0$. We will divide the estimation into two parts:

Lower Bound

Let x be in the support of μ so we have $f(x) > 0$. In this section we aim to establish a lower bound of $F_{\rho, \mu}(x, \sigma)$ that goes to infinity as we increase σ . Since $\lim_{\sigma \rightarrow 0} F_{\rho, \mu}(x, \sigma) = \log \rho <$

0, this will prove the existence of σ^* . Assume μ has C^1 density f and $f, \|\nabla f\|$ are bounded. Recall

$$Z := \int_0^\infty \exp(-w(t))dt < \infty. \quad (3.2)$$

Consider

$$\begin{aligned} \int \exp(-w(\sigma^2 \|x - x'\|^2))d\mu(x') &= \int \exp(-w(\sigma^2 \|x - x'\|^2))f(x')dx' \\ &= \left(\frac{Z}{\sigma}\right)^d \int \frac{\exp(-w(\sigma^2 \|x - x'\|^2))}{(Z/\sigma)^d} f(x')dx'. \end{aligned} \quad (3.3)$$

Because $\lim_{x \rightarrow \infty} w(x) = \infty$, the integrand of the above integral converges weakly to $\delta_x f(x')$. This can be easily proved using truncation. **[Yi: I discussed with several people doing analysis and I am positive that this is all we need for the integrand to converge to Dirac Delta. Correct me if I am wrong.]** Along with the fact f is bounded, there exists some $\epsilon(\sigma) \rightarrow 0$ as $\sigma \rightarrow \infty$ and a constant $C > 0$, that

$$\begin{aligned} -\log \left(\int \exp(-w(\sigma^2 \|x - x'\|^2))d\mu(x') \right) &= -d \log \left(\frac{Z}{\sigma} \right) - \log f(x) + \epsilon(\delta) \\ &\geq C \log \sigma - \log f(x). \end{aligned} \quad (3.4)$$

Next, since f is bounded, there exists $M > 0$ such that $f \leq M$. Let T be the unique positive real number with $\log(T) = T/2$. Then for each $t > T$, we have $\log(t) - t \leq t/2$. This means

$$\begin{aligned} &\int \frac{w(\sigma^2 \|x - x'\|^2) \exp(-w(\sigma^2 \|x - x'\|^2))}{\int \exp(-w(\sigma^2 \|x - x'\|^2))d\mu(x')} d\mu(x') \\ &\leq M \int \frac{w(\sigma^2 \|x - x'\|^2) \exp(-w(\sigma^2 \|x - x'\|^2))}{\int \exp(-w(\sigma^2 \|x - x'\|^2))f(x')dx'} f(x')dx' \\ &\leq M \int_{w \leq T} \frac{w(\sigma^2 \|x - x'\|^2) \exp(-w(\sigma^2 \|x - x'\|^2))}{\int \exp(-w(\sigma^2 \|x - x'\|^2))f(x')d(x')} f(x')dx' \\ &\quad + M \int_{w > T} \frac{\exp(-w(\sigma^2 \|x - x'\|^2)/2)}{\int \exp(-w(\sigma^2 \|x - x'\|^2))f(x')d\mu(x')} dx' \\ &\leq M'(x) + \left(\frac{Z'}{Z}\right)^d \frac{1}{f(x)} \equiv M''(x) < \infty \end{aligned} \quad (3.5)$$

for some constant $M'(x) > 0$ if we assume

$$Z' := \int \exp(-w(t)/2)dt < \infty. \quad (3.6)$$

Combine (3.1), (3.4) and (3.5) and we get

$$F_{\rho, \mu}(x, \sigma) \geq C \log \sigma - \log f(x) + \log \rho. \quad (3.7)$$

Monotonicity

We take derivative of $F_{\rho, \mu}(x, \sigma)$ with respect to σ using (3.1). To make the computation clearer, we use simplified notation in the first several steps:

$$w = w(\sigma^2 \|x - x'\|^2); \quad \exp(\cdot) = \exp(-w(\sigma^2 \|x - x'\|^2)). \quad (3.8)$$

Omit $d\mu(x')$ in the integration and we have

$$\begin{aligned}
\frac{\partial}{\partial \sigma} F_{\rho, \mu}(x, \sigma) &= - \int \frac{w' \exp(\cdot)}{\int \exp(\cdot)} - \int \frac{w \exp(\cdot)'}{\int \exp(\cdot)} + \int \frac{w \exp(\cdot) (\int \exp(\cdot))'}{(\int \exp(\cdot))^2} - \frac{(\int \exp(\cdot))'}{\int \exp(\cdot)} \\
&= - \int \frac{w \exp(\cdot)'}{\int \exp(\cdot)} + \int \frac{w \exp(\cdot) (\int \exp(\cdot))'}{(\int \exp(\cdot))^2} \quad (\text{The first and fourth terms cancel.}) \\
&= 2\sigma \left(\int \frac{w \cdot w' \cdot \|x - x\|^2 \cdot \exp(\cdot)}{\int \exp(\cdot)} - \frac{\int w \cdot \exp(\cdot) \int w' \cdot \|x - x\|^2 \exp(\cdot)}{(\int \exp(\cdot))^2} \right). \tag{3.9}
\end{aligned}$$

We are able to move differentiation inside the integral because of dominant convergence theorem. To show that $\frac{\partial}{\partial \sigma} F_{\rho, \mu}(x, \sigma) \geq 0$, we just need to show

$$\int w \cdot w' \cdot \|x - x\|^2 \cdot \exp(\cdot) \int \exp(\cdot) \geq \int w \cdot \exp(\cdot) \int w' \cdot \|x - x\|^2 \exp(\cdot). \tag{3.10}$$

To tackle this inequality, we need the following lemma:

Lemma 1 (Theorem 7, [2]). *Let $f(x), g(x), h(x)$ be continuous functions on $[a, b]$ such that*

$$(g(x) - g(y)) \left(\frac{f(y)}{h(y)} - \frac{f(x)}{h(x)} \right) \geq 0. \tag{3.11}$$

Then the inequality

$$\frac{\int_a^b f(x) dx}{\int_a^b h(x) dx} \geq \frac{\int_a^b f(x) g(x) dx}{\int_a^b h(x) g(x) dx} \tag{3.12}$$

holds. If (3.11) reverses, then (3.12) reverses.

The lemma can easily be extended to integral over $\mathbb{R}$ if every integral converges, which is our case. For each $x, y, x' \in \mathbb{R}$ if the equality

$$(w'(\sigma^2 \|x - x'\|^2) - w'(\sigma^2 \|y - x'\|^2)) \left(\frac{1}{w(\sigma^2 \|y - x'\|^2)} - \frac{1}{w(\sigma^2 \|x - x'\|^2)} \right) \geq 0 \tag{3.13}$$

holds, then by letting $f(x) = \exp(\cdot)$, $h(x) = w \cdot \exp(\cdot)$ and $g(x) = w' \cdot \|x - x\|^2$, (3.10) holds. **[Yi: It seems that by (3.13), w has to be a convex function. But we don't have to put $\|x - x'\|^2$ inside w , so there is room for improvement.]**

Wrap Up

We know that $\lim_{\sigma \rightarrow 0} F_{\rho, \mu}(x, \sigma) = \log \rho < 0$ and by 3.7 $\lim_{\sigma \rightarrow \infty} F_{\rho, \mu}(x, \sigma) = \infty$. Moreover, since $F_{\rho, \mu}(x, \sigma)$ is monotone increasing in σ , we can conclude there exists a unique $\sigma^* = \sigma_{\rho, \mu}^*(x)$ for all $\rho \in (0, 1)$ and $x \in \mathbb{R}^d$. **[Yi: TODO: Improve the power of $\|x - x'\|$ inside w .]**

References

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